



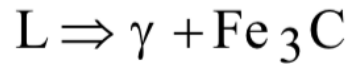
STM Exercise 8

Phase Diagrams
(Fe-C)

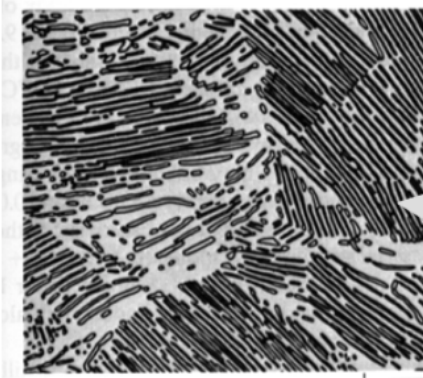
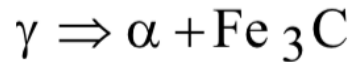
IRON-CARBON (Fe-C) PHASE DIAGRAM

- 2 important points

-Eutectic (A):

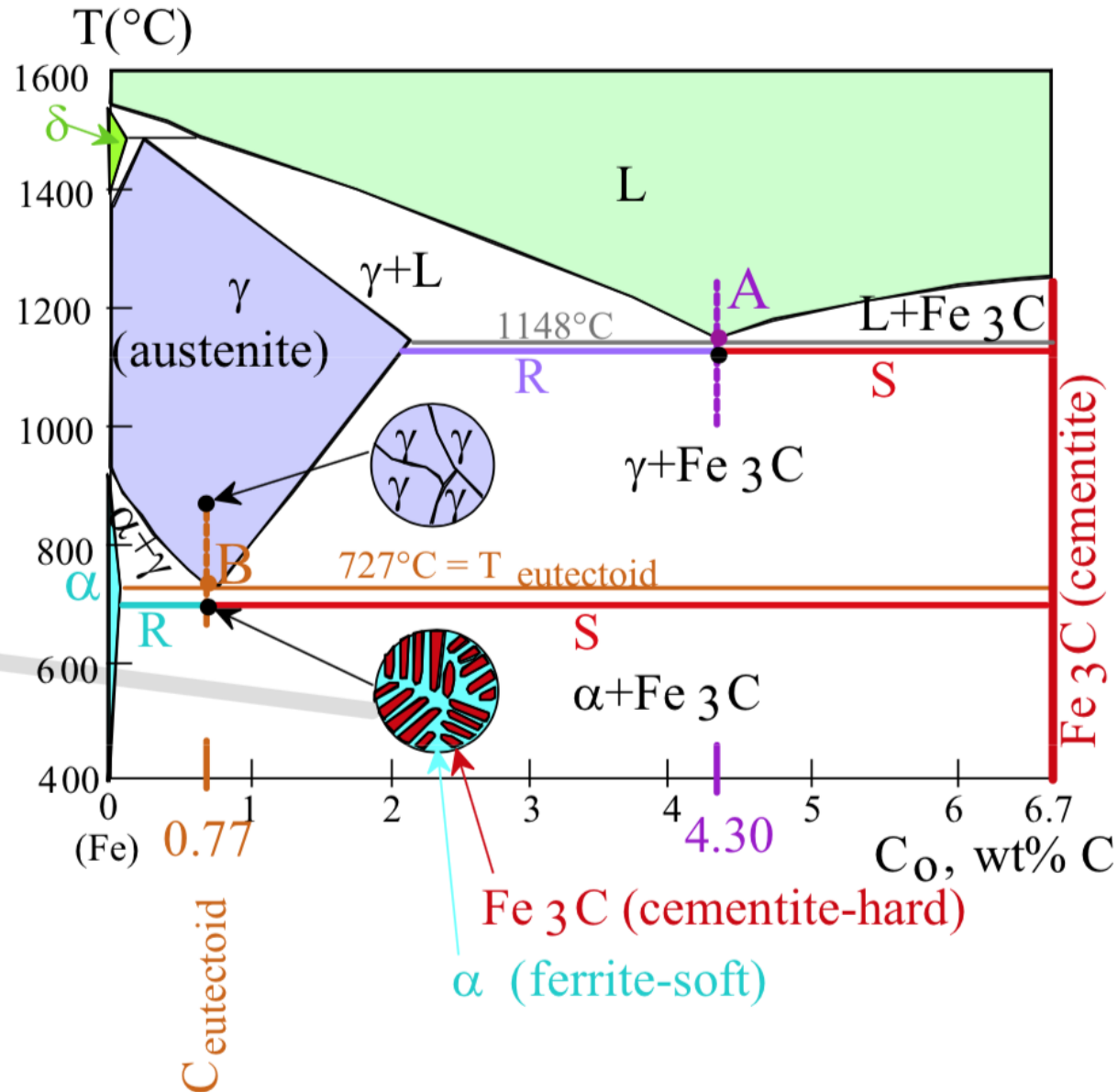


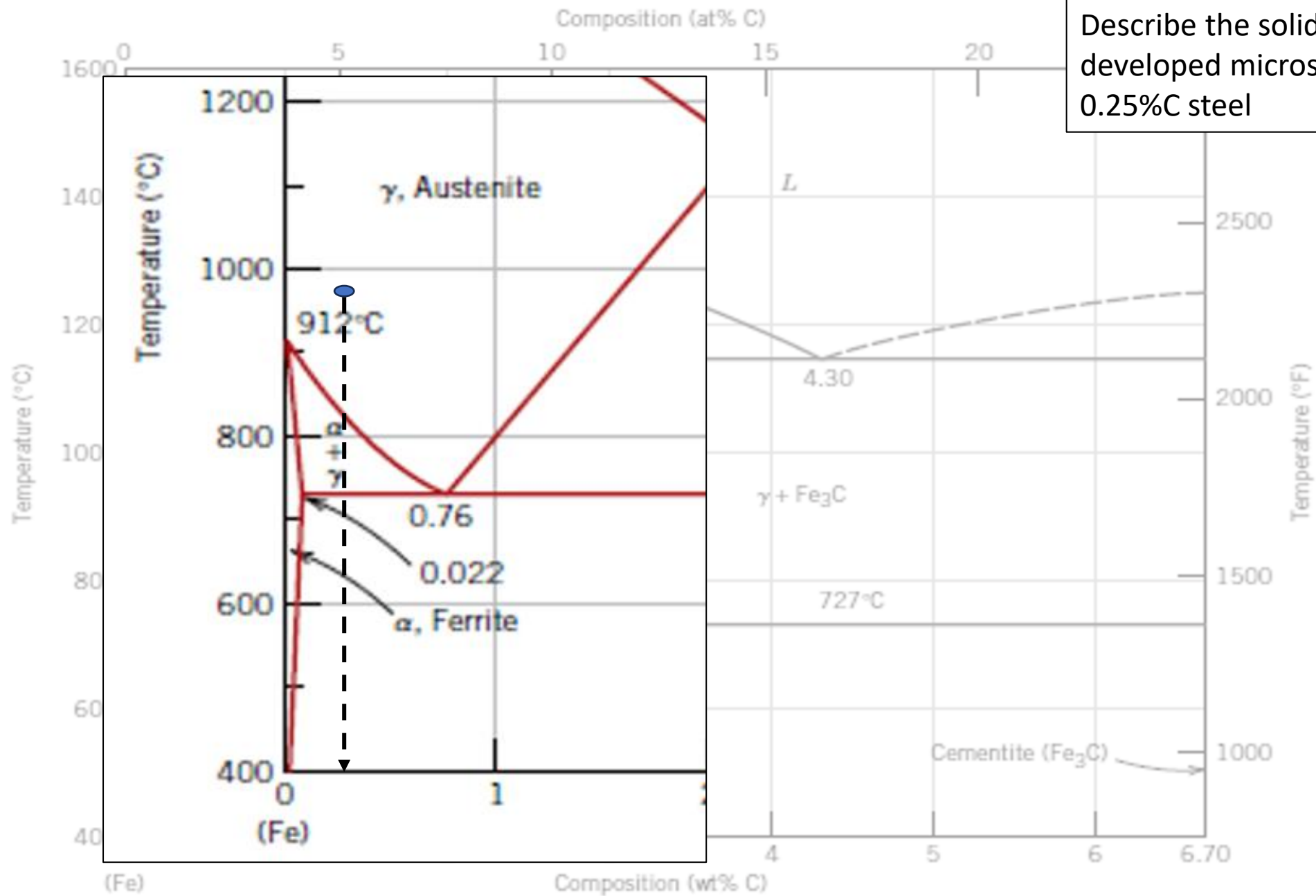
-Eutectoid (B):



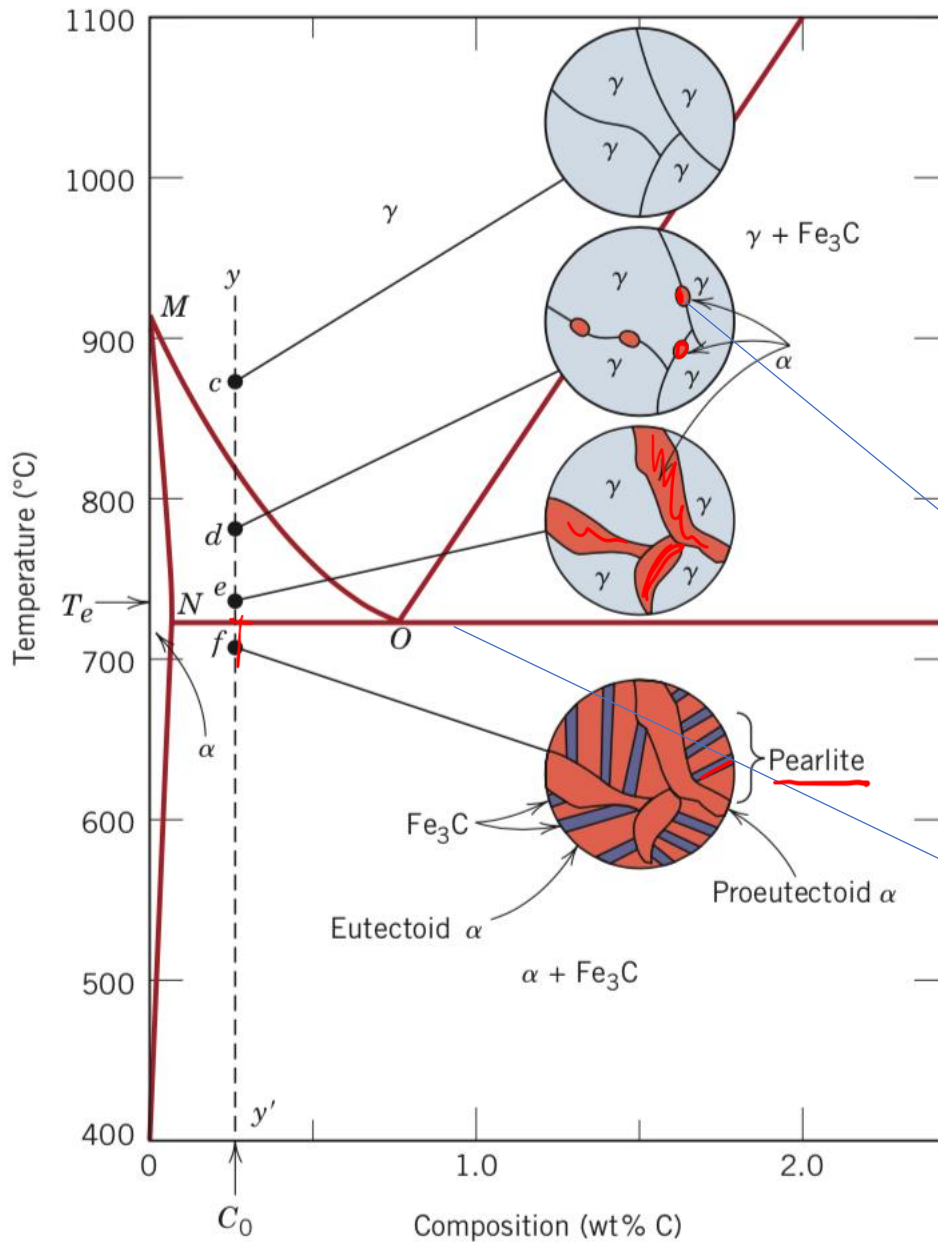
120 μm

Result: Pearlite = alternating layers of α and Fe_3C phases.





Describe the solidification (evidencing developed microstructures) of a 0.25%C steel



Main Phases

Ferrite (α) → Solid solution of carbon in α -iron (body-centered cubic, BCC).

- Carbon content (maximum 0.022 wt% C).

Austenite (γ) → Solid solution of carbon in γ -iron (face-centered cubic, FCC).

- Can contain up to 2.14 wt% C.
- Stable between approximately 727°C and 1493°C.

Cementite (Fe_3C) → Hard and brittle intermetallic compound.

- Contains 6.67 wt% C

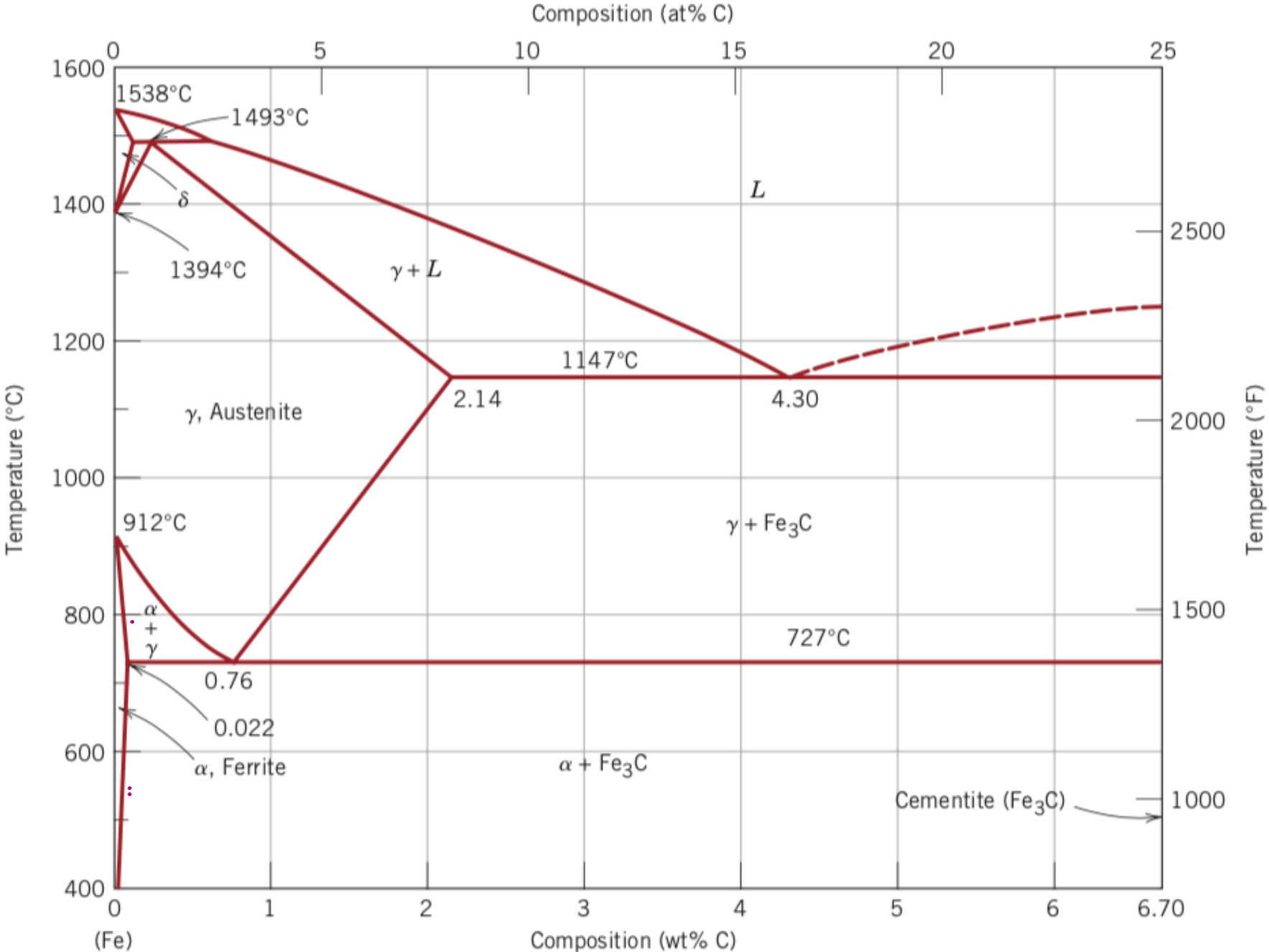
α Proeutectoid alpha(ferrite)

727°C Eutectoid reaction: Austenite → Ferrite + Cementite

Pearlite

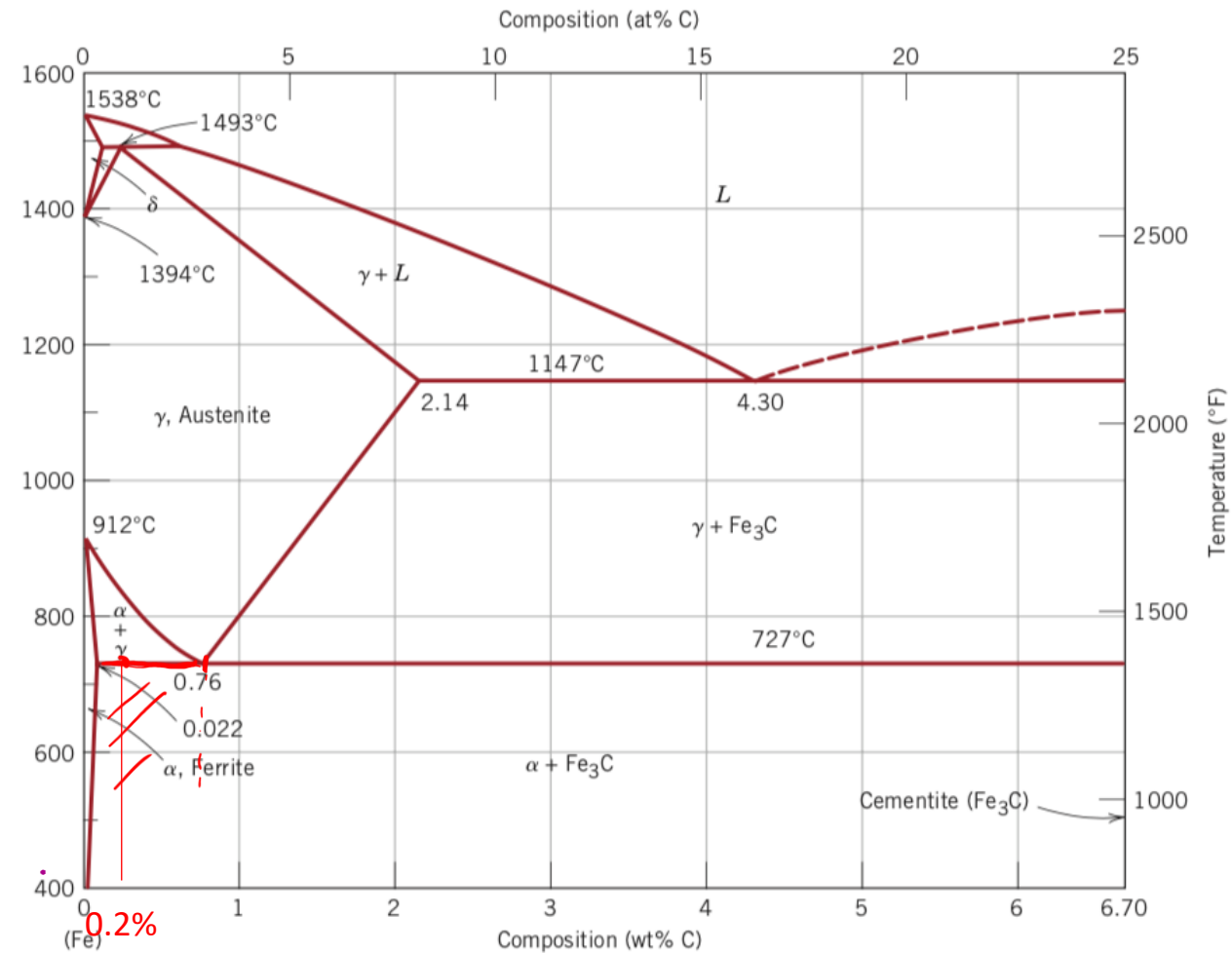
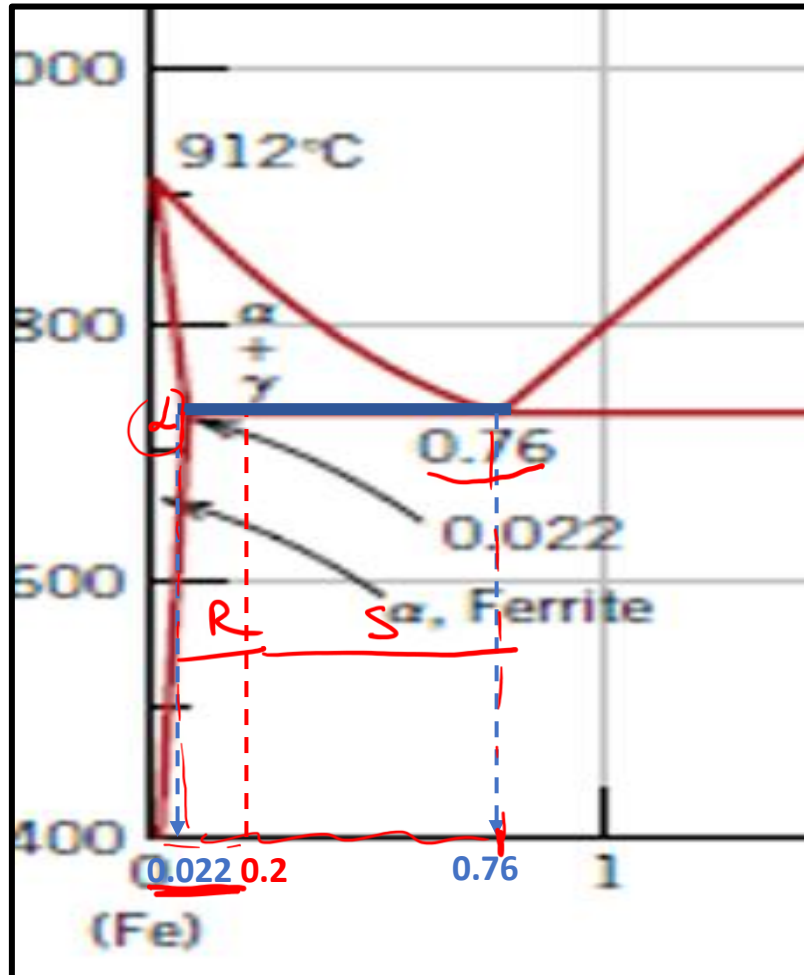
Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel



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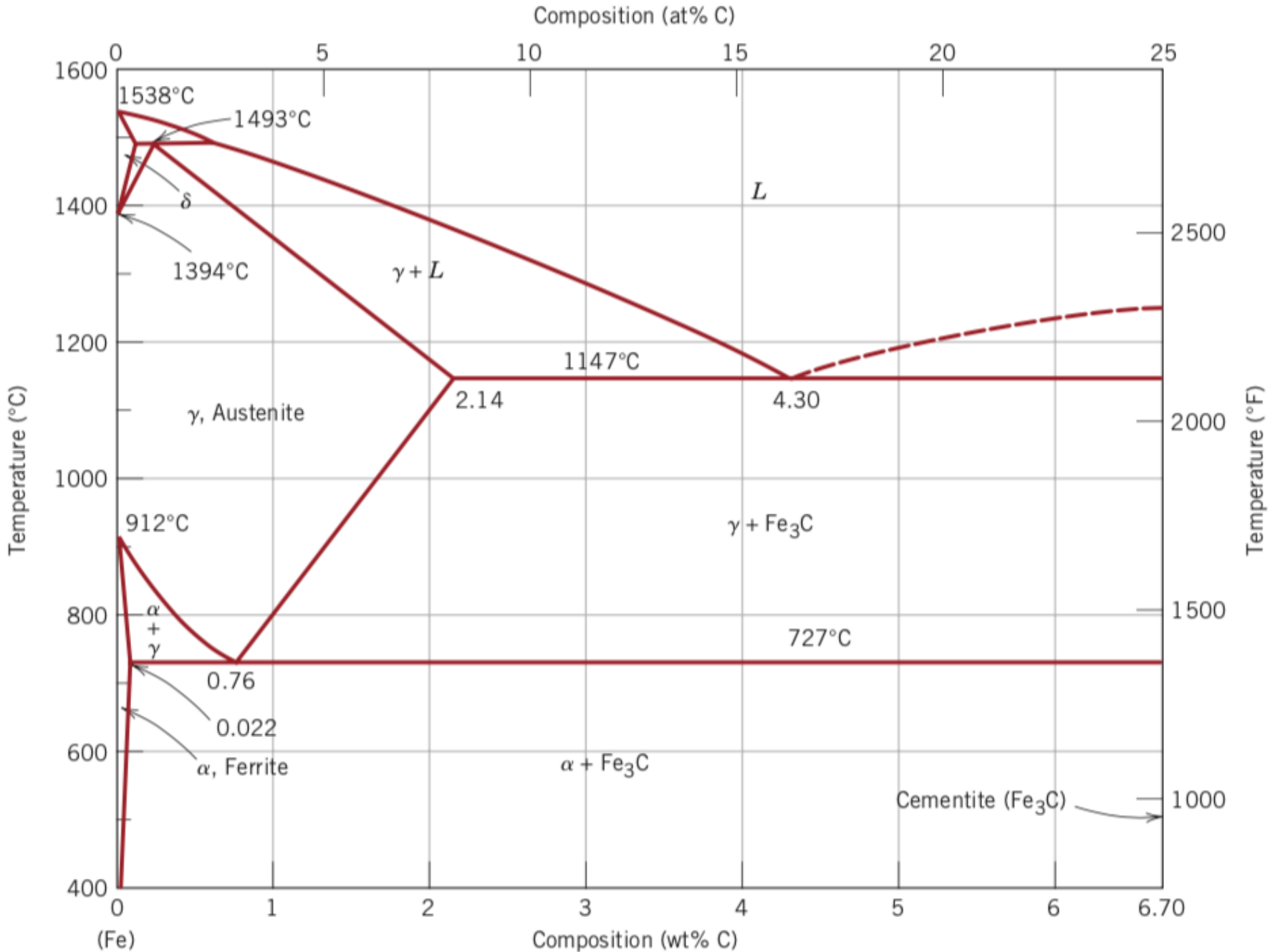


Lever rule on the eutectoidic tie line (blue line in figure):

$$\% \alpha = \frac{(0.76 - 0.2)}{(0.76 - 0.022)} \times 100 = 76\%$$

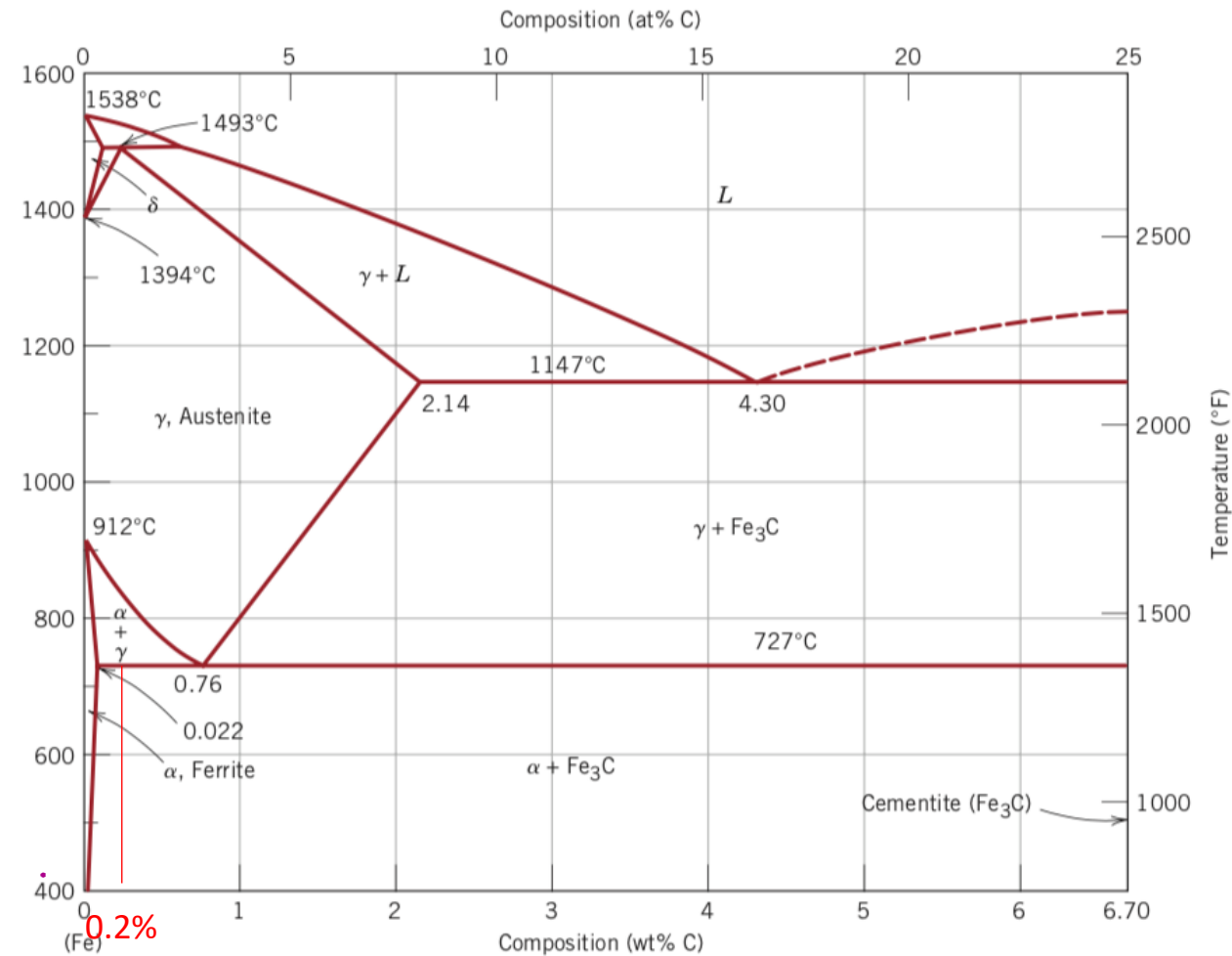
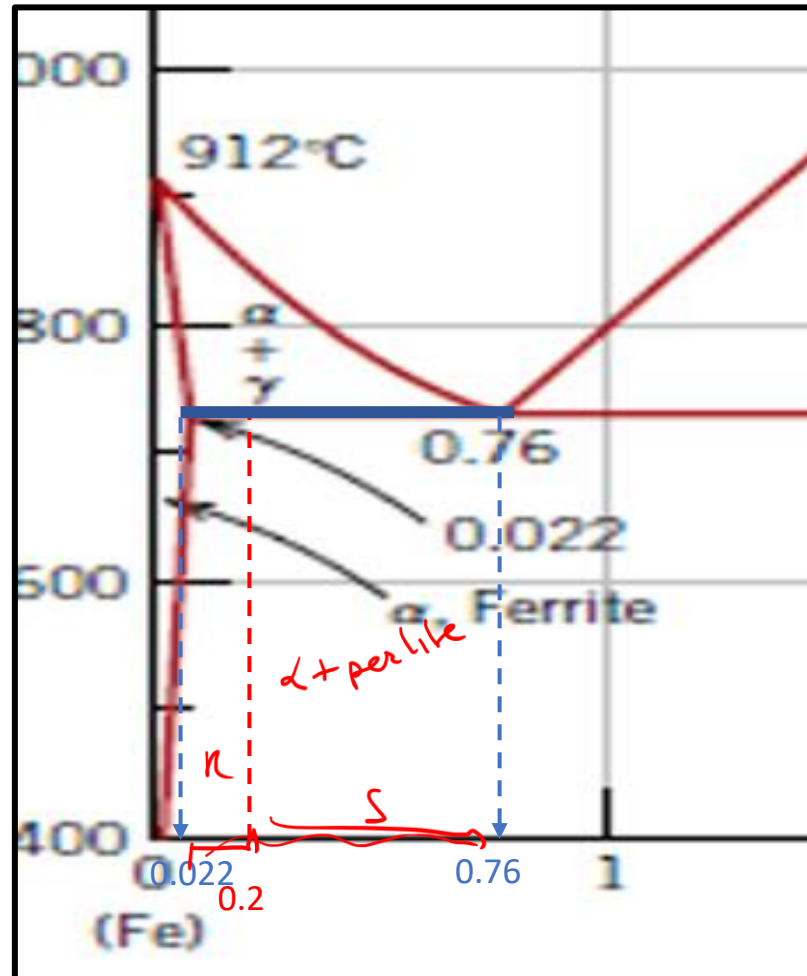
Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel



Use the Fe-Fe₃C phase diagram to calculate:

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Lever rule on the eutectoid tie line (blue line in figure):

$$\therefore \% \gamma = \frac{(0.2 - 0.022)}{(0.76 - 0.022)} \times 100 = 24\%$$

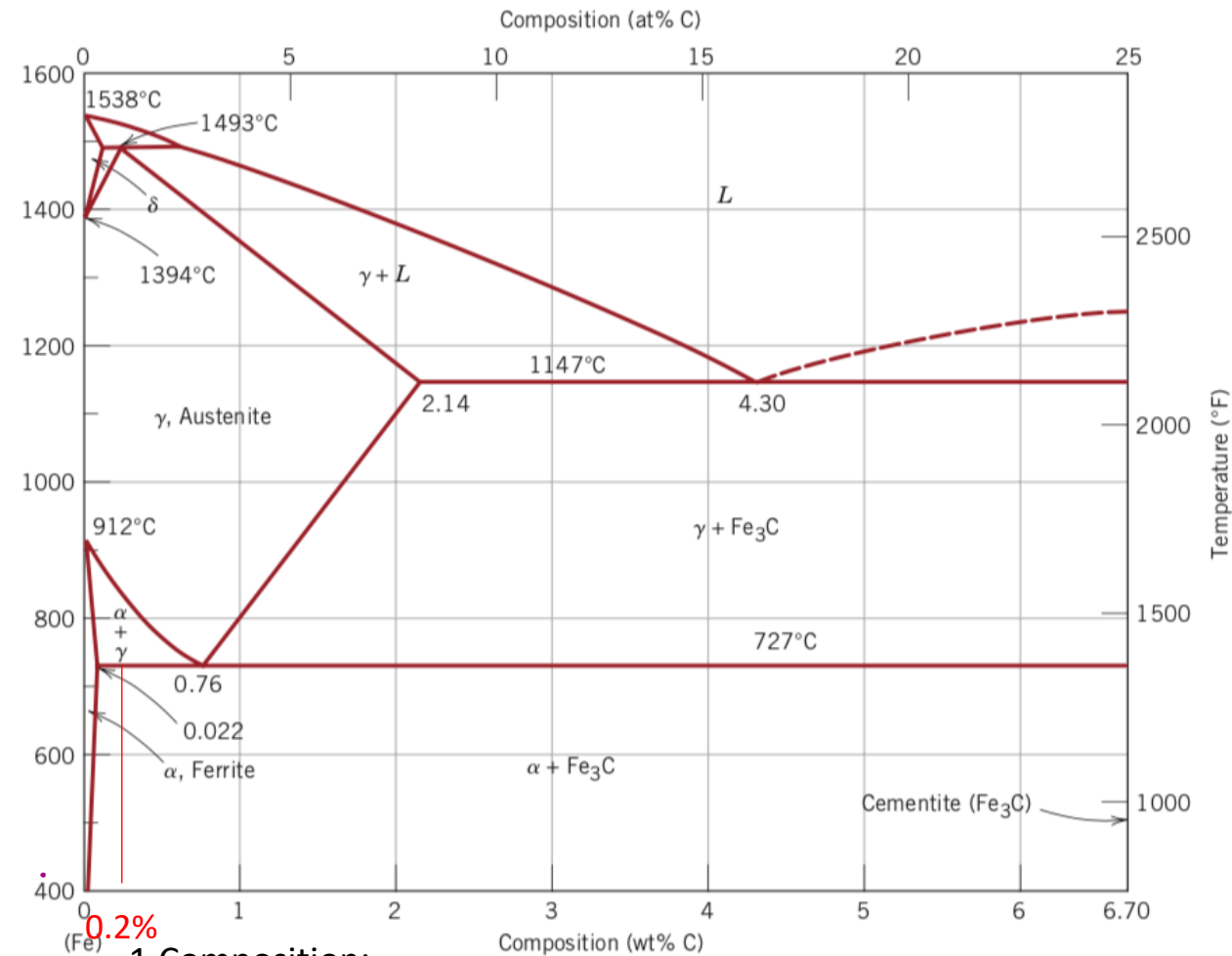
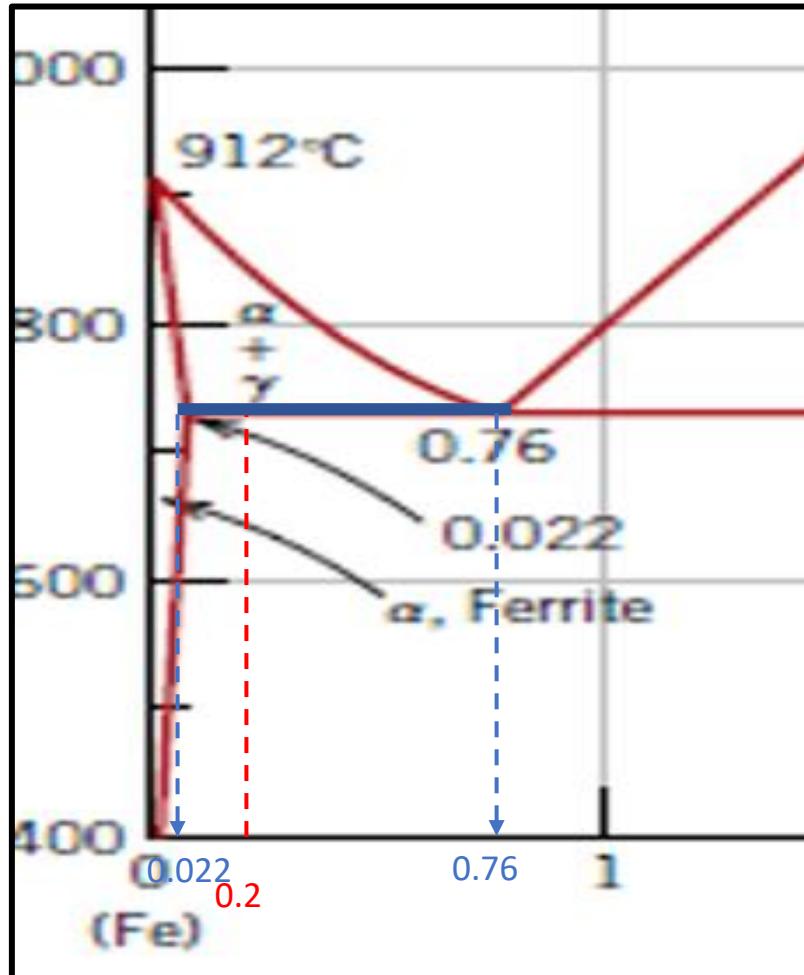
The % of pearlite is equal to the one of austenite just before the eutectoid

At **727°C**, austenite (γ) of eutectoid composition (0.76 wt% C) transforms into ferrite (α) and cementite (Fe_3C) according to:

$\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$ we are into the reaction

Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel



1.Composition:

1. Go to the x-axis (wt% C).
2. Find 0.2 wt% C and draw a vertical line up from there.

Pearlite occurs at eutectoid temperature, about 727 °C.

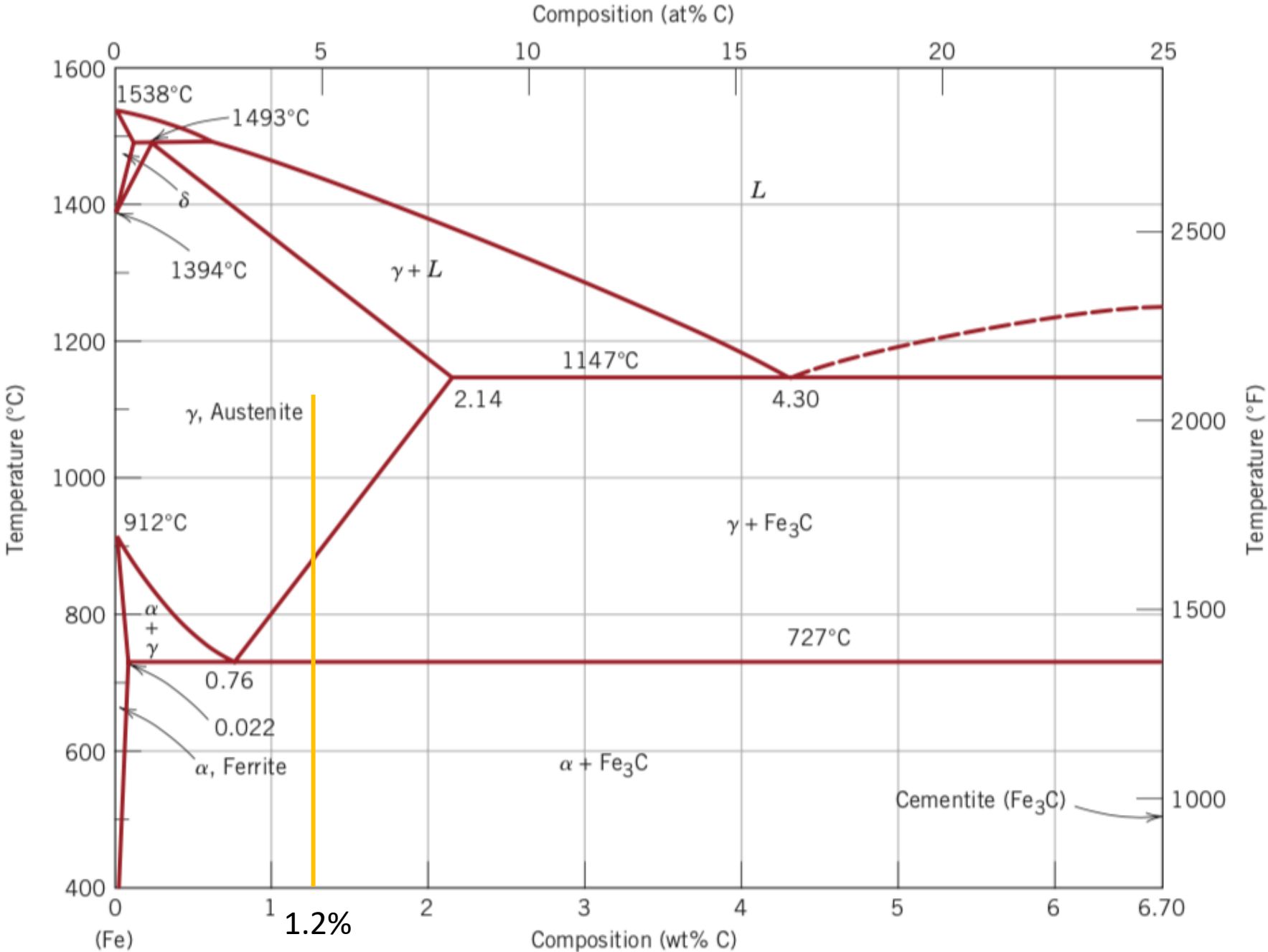
1. So draw a horizontal line at 727 °C

2.Intersection:

1. The point where 0.2 wt% C and 727 °C meet is where you apply the lever rule.
2. Just below 727 °C the structure is:
proeutectoid ferrite + pearlite.

Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of pearlite for a 1,2 % C steel



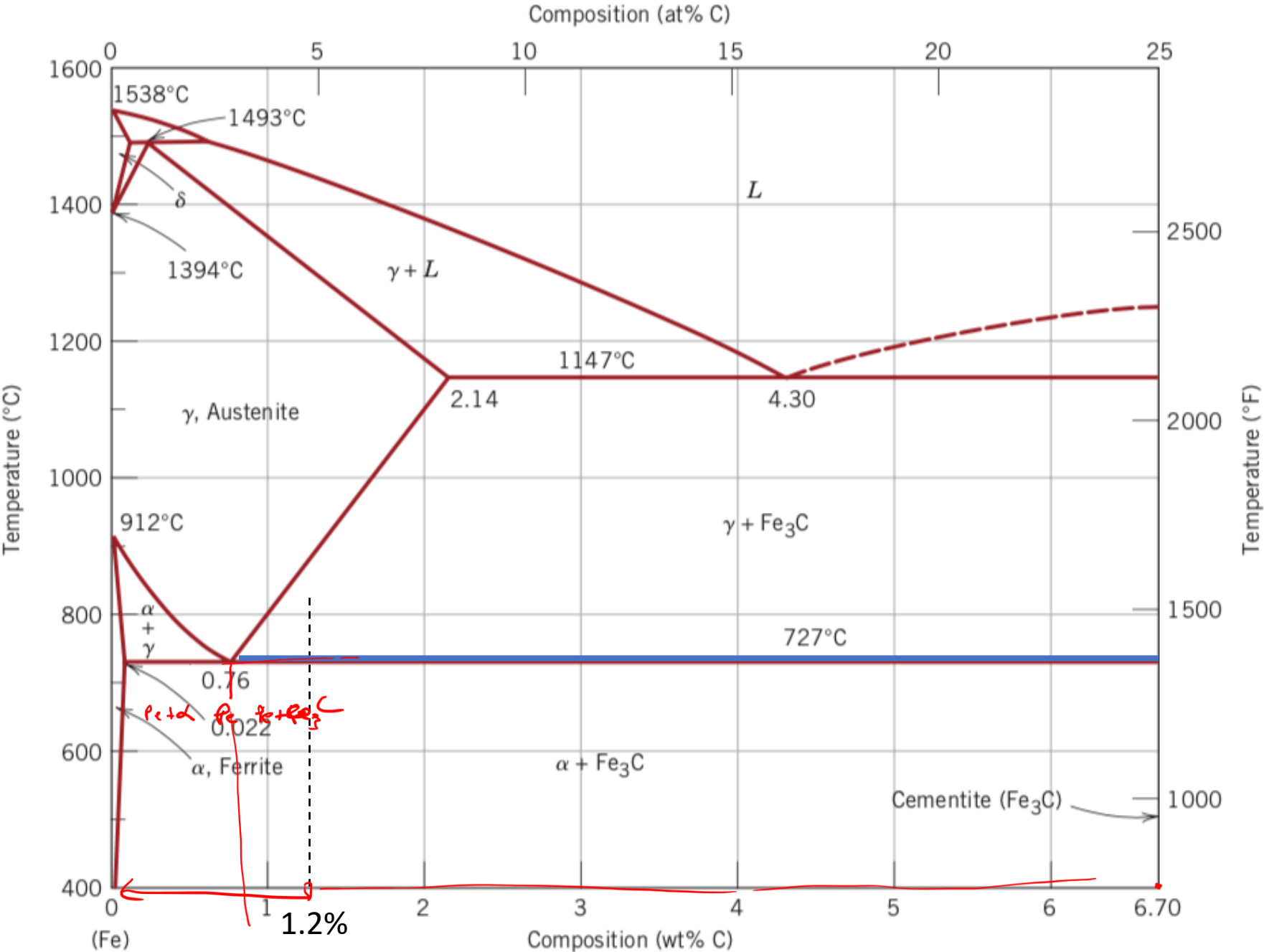
Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of pearlite for a 1,2 % C steel

During eutectoidic transformation all the austenite (γ) is transformed in α +Fe₃C pearlitic structure.

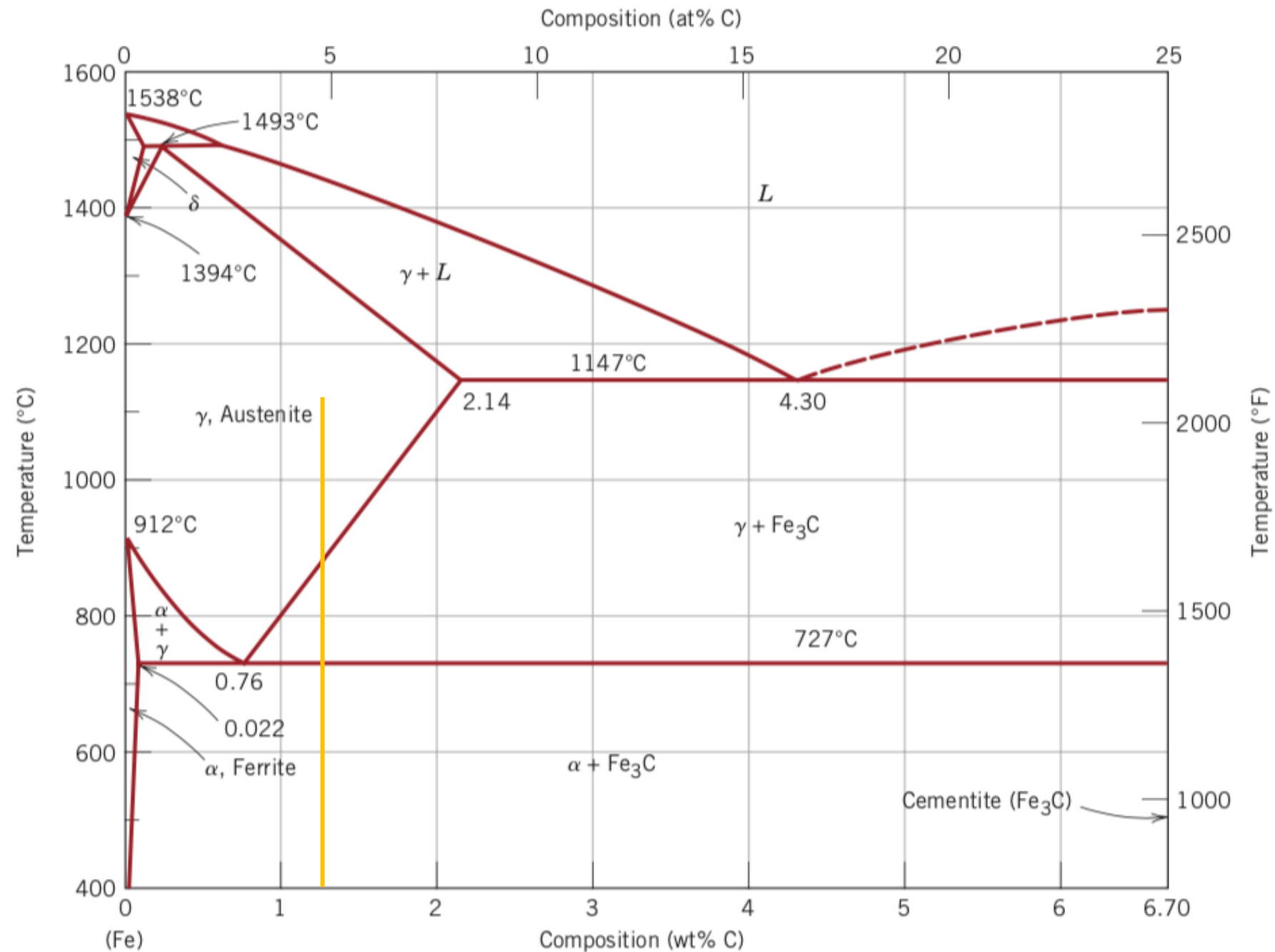
So, the % of pearlite is equal to the % of γ at the beginning of the eutectoidic transformation

$$\% \gamma (723^{\circ}\text{C}) = \frac{(6.70 - 1.2)}{(6.70 - 0.76)} \times 100 = 92.5\%$$



Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel



Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel

We find the cementite primary → so we need to go at 727°C

Lever rule at 727°C
 $\%Fe_3C \text{ (pro-eutectoid)} = \frac{(1.2 - 0.76)}{(6.70 - 0.76)} * 100 = 7.4\%$

